

Course Code: CS2404

Name of the Programme: B.Tech(H)

Name of the Course: IoT

Semester: 3rd

Course credit: 3

**No. of hours per week :
(L + T + P) 0:2:2**

**Total no. of Teaching hours
: 60 Hours**

Course Objectives:

- a) To introduce the fundamental concepts of the Internet of Things (IoT), including its architecture, ecosystem, and real-world applications.
- b) To enable students to interface and program microcontrollers such as Arduino and ESP32 with various sensors and actuators.
- c) To provide hands-on experience in connecting IoT devices to cloud platforms using protocols like MQTT and HTTP for real-time data communication.
- d) To develop the ability to build, test, and evaluate IoT applications involving data acquisition, cloud integration, and basic analytics.

Course outline (Syllabus of the course)

Syllabus:

Module – 1: Introduction to IoT and Development Platforms	No. of hours
	9
Introduction to the concept, origin, and evolution of the Internet of Things (IoT). It covers the architecture of IoT systems, the significance of IoT in today's connected world, and its application across various domains such as smart homes, agriculture, and healthcare. Students are introduced to development boards like Arduino UNO and ESP32 , their programming environments (Arduino IDE), and the workflow of writing, compiling, and uploading code. Basic GPIO operations, digital input/output, and LED control serve as the foundation for hands-on learning.	

Module – 2: Sensors and Actuators – Interfacing and Control	No. of hours
	9
This module focuses on understanding the types, working principles, and integration of sensors and actuators. Students learn to interface analog and digital sensors with Arduino/ESP32 and write programs to read sensor data. Actuators such as buzzers, motors, and relays are also introduced. Students work on real-time data acquisition and feedback mechanisms, such as triggering alarms or controlling devices based on sensor inputs.	

Module – 3: Networking and IoT Protocols	No. of hours
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Networking fundamentals and how IoT nodes communicate using wireless technologies like Wi-Fi and Bluetooth. It introduces communication protocols such as HTTP, MQTT, and Blynk, and guides students through connecting devices to cloud platforms. This module emphasizes sending and receiving real-time sensor data over the internet, accessing dashboards, and remotely controlling devices through smartphones or web interfaces.	

Module – 4 : Cloud Integration and Data Handling	No. of hours
	9
Concept of cloud-based data storage, retrieval, and visualization. Topics include real-time data monitoring, pushing sensor data to platforms like ThingSpeak and visualizing trends using dashboards. The module also discusses time-series data, structured vs. unstructured data, and introduces basic analytics. Interfacing with online databases like Google Sheets or MySQL via ESP32 is demonstrated to showcase how IoT supports data-driven decisions.	

Module – 5: IoT Applications and Mini Project	No. of hours
	9
Focuses on the design, integration, and implementation of complete IoT applications. It encourages students to combine knowledge from previous modules to build real-world systems such as smart lighting, weather stations, or home automation setups. Project planning, debugging, and performance evaluation are emphasized. Students also explore basic security practices and optimization for power and performance in IoT systems.	

Course outcomes
a) Understand the core principles of IoT systems and identify their applications across various domains.
b) Interface and program sensors, actuators, and communication modules with Arduino and ESP32 boards.
c) Implement network communication between IoT devices and cloud platforms using industry-standard protocols.
d) Design and demonstrate end-to-end IoT solutions integrating hardware and cloud components with basic data visualization.

Text Books	
1	Arshdeep Bahga & Vijay Madisetti – <i>Internet of Things: A Hands-On Approach</i>
2	Agus Kurniawan -Learning ESP32 using Arduino IDE , Publisher: PE Press

Reference Books	
1	Simon Monk-Programming Arduino: Getting Started with Sketches (2nd Edition), McGraw-Hill Education
2	Raj Kamal – <i>Internet of Things: Architecture and Design Principles</i>

Lab Activities to be done	
Sl. No.	Lab Activities
1	LED Blinking and GPIO Control using Arduino and ESP32
2	Reading Analog Sensor Data (LM35/Photoresistor) and Displaying via Serial Monitor
3	Interfacing Ultrasonic Sensor for Distance Measurement
4	Controlling a Stepper Motor or Buzzer using Relay Module with Sensor Trigger
5	Interfacing DHT11 Sensor and Displaying Temperature/Humidity on LCD or OLED
6	Wi-Fi Connectivity with ESP32 and Sending Data to ThingSpeak
7	Implementing MQTT Protocol for Sensor Data Communication using ESP32
8	Remote Control of Devices via Blynk App or Firebase
9	Logging Sensor Data to Google Sheets using ESP32
10	Mini Project: Smart Environment Monitoring / Home Automation System